

Rafael Mendonça Duarte

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Education

Bachelor of Science in Information Systems, University of São Paulo (USP) – Jan 2025 – Dec 2028 (Expected)
ICMC
Relevant Coursework: Data Structures, Modern Operating Systems, Object-Oriented Programming (OOP), Calculus, Linear Algebra, Statistics.

Experience

Research Intern (Fully Funded – FAPESP, Grant #2026/16921-8), University of São Paulo August 2025 – Present

- Currently investigating graph fusion strategies and representative/synthetic vertices to improve the structural representation of Graph Convolutional Networks (GCNs) for image classification.
- Previously (Nov 2025 – Jul 2026), **first-authored** "*Gaussian Rank-Based Neighborhood Degree for GNNs in Image Classification*" (**SBBB 2026**), achieving **State-of-the-Art (SOTA)** performance and **improving accuracy by up to 14%** over previous baselines.

Key Skills: Graph Neural Networks (GNNs), PyTorch, Machine Learning, Data Structures and Data Science.

Director of High-Performance Computing (HPC) Study Group, IEEE Computer Society Chapter (USP) Jun 2025 – Present

- Orchestrated a **performance and scalability evaluation** of databases on **resource-constrained ARM hardware** (TV Box) utilizing Docker environments.
- Analyzed latency and throughput trade-offs across different architectures, benchmarking direct DB access with focus to **optimize I/O bottlenecks**.

Key Skills: Docker, Database Architecture, Performance Benchmarking, System Administration and FastAPI.

Digital Literacy Instructor for Seniors, "Informática para a Terceira Idade" February 2026 – June 2026

- Facilitated **digital inclusion** classes for the elderly community, teaching essential computer skills and internet navigation to promote **technical autonomy**.

Awards & Extracurricular Activities

Scientific Olympiads Medalist & Finalist, National Competitions 2021 – 2024

- Achieved **medals** in the Kangaroo Math Competition, Brazilian Astronomy Olympiad (OBA), and MOB-FOG; two-time finalist (4th phase) at the National History Olympiad (ONHB).

Model United Nations (MUN) Awardee, FAMUN & CJONU 2022 – 2023

- Awarded "**Best Position Paper**" and "**Best Director**", highlighting strong communication, diplomacy, and research skills.

Technical Skills

Programming & Web: Python, Java, SQL.

Machine Learning & Data Science: PyTorch, Graph Neural Networks, NLP (Word2Vec), XGBoost, Pandas.

Tools & Infrastructure: Docker, Git, Linux.

Core Concepts: Data Structures, Object-Oriented Programming (OOP), Database Architecture, Statistics.

Languages

Portuguese: Native | **English:** Intermediate — proficient in technical reading and writing